

Orange juice improved lipid profile and blood lactate of overweight middle-aged women subjected to aerobic training.

OBJECTIVE: This study investigated how consumption of orange juice associated with aerobic training affected serum lipids and physical characteristics of overweight, middle-aged women. **METHODS:** The experimental group consisted of 13 women who consumed 500 mL/d of orange juice and did 1h aerobic training 3 times a week for 3 months. The control group consisted of another 13 women who did the same aerobic training program but did not consume orange juice. **RESULTS:** At the end of the experiment, the control group lost an average of 15% of fat mass ($P<0.05$) and 2.5% of weight ($P<0.05$), whereas the experimental group lost 11% of fat mass and 1.2% of weight ($P<0.05$). Consumption of orange juice by the experimental group was associated with increased dietary intake of vitamin C and folate by 126% and 61% respectively. Serum LDL-C decreased 15% ($P<0.05$) and HDL-C increased 18% ($P<0.05$) in the experimental group, but no significant change was observed in the control group. Both groups improved the anaerobic threshold by 20% ($P<0.05$), but blood lactate concentration decreased 27% in the experimental group compared to the 17% control group, suggesting that experimental group has less muscle fatigue and better response to training. **CONCLUSIONS:** The consumption of 500 mL/d of orange juice associated with aerobic training in overweight women decreased cardiovascular disease risk by reducing LDL-C levels and increasing HDL-C levels. This association also decreased blood lactate concentration and increased anaerobic threshold, showing some improvement in the physical performance. Copyright © 2010 Elsevier Ireland Ltd. All rights reserved.